

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester III Examination, 2023-24

Botany

Paper: BOB-301 — Plant Ecology and Physiology

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions, selecting two each from Sections A and B and Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer/define the following in not more than 50 words each: [10 × 1 = 10]

- (a) Invasive species
- (b) Habitat Mutualism
- (c) Density
- (d) Ecological pyramid
- (e) Allogenic succession
- (f) Apical dominance
- (g) Hydathodes
- (h) Embolism
- (i) H⁺-ATPase
- (j) Emerson enhancement effect

Section-A

Question 2. Define population. Give a detailed account of various characteristics of a population. [15]

Question 3. Write descriptive notes on any two of the following: [2 × 7.5 = 15]

- (a) General soil profile
- (b) Universal model of energy flow
- (c) Raunkiaer's life forms

Question 4. Write critical notes on any three of the following: [3 × 5 = 15]

- (a) Adaptations in Xerophytes
- (b) Analytical characters of community

- (c) Grazing and detritus food chain
- (d) Phosphorus cycle

Question 5. What is ecological succession? Describe the various sequential stages of a typical hydrosere. [15]

Section-B

Question 6. Give an account of the C₄ and CAM pathways of CO₂ fixation in plants. [15]

Question 7. Write critical notes on any two of the following: [2 × 7.5 = 15]

- (a) Nitrate assimilation in plants
- (b) Photophosphorylation
- (c) Photorespiration

Question 8. Write short notes on any three of the following: [3 × 5 = 15]

- (a) Membrane transport
- (b) Phloem loading
- (c) Structure of phytochrome
- (d) Electron transport system in mitochondria

Question 9. Define growth. Differentiate between growth and development. Discuss the physiological roles of auxins and gibberellins in plants. [15]

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